

Violence Reduction and Access to Higher Education: Post-Conflict Evidence from Colombia

MY400 Seminar Group 4 - Candidate number: 61296

January 2026

1 Introduction

A substantial body of academic evidence shows that education—viewed as an investment in human capital—plays a crucial role in the promotion of development, economic growth, and social mobility by raising income levels and improving labour productivity (Becker 2002; Mincer 1958). Academic research consistently finds that returns to higher education are both significant and increase over time (Psacharopoulos and Patrinos 2018). However, human capital accumulation is shaped not only by individual choices and abilities, but also by broader social, economic, political, and cultural conditions, among other contextual factors (Attanasio 2015).

Countries that have experienced periods of armed conflict and violence, such as Iraq, Bosnia, Peru, Rwanda, and several Sub-Saharan nations, often face substantial negative effects on human capital formation (Jürges et al. 2022; Diwakar 2015; Swee 2015; Leon 2012; Akresh 2008; Justino 2010). These impacts tend to be even more severe when conflicts are highly intense and occur in contexts with weak state institutions (Unfried and Kis-Katos 2023). The consequences are commonly assessed through indicators such as school enrollment, educational attainment, dropout rates, and academic performance across primary, secondary, and higher education levels.

These negative effects arise through the direct impact of violence on education supply, including financial constraints that limit public goods provision, damage to educational infrastructure, threats to teachers, increases in the opportunity cost of studying, and other related factors (Montaño-Bardales and Palacios 2025; Milton and Barakat 2018; Rodriguez and Sanchez 2012; Sánchez Torres and Díaz Escobar 2005; Collier et al. 2003). Colombia is not an exception. The country experienced an internal armed conflict for more than fifty years, and empirical evidence documents its negative effects on key educational outcomes, including enrollment, retention, school attendance, and graduation

rates at both the primary and secondary levels (Soler 2016, Fergusson, Ibáñez, and Riano 2020, Montaña-Bardales and Palacios 2025).

Despite all evidence regarding the effects of conflict and violence on human capital formation, there is less evidence of the impact of violence reduction. Through an analysis of cross-sections of countries, Chen, Loayza, and Reynal-Querol 2008 found that transitions to peace can lead to educational recovery. However, Lai and Thyne 2007 examined the effects of post-civil war environment on educational expenditures and primary, secondary, and tertiary education, finding that post-conflict periods are associated with reductions in educational expenditure and enrollment rates for all levels.

For Colombia, recent literature has increasingly examined the effects of reduced violence following the 2016 peace agreement signed between the guerrilla group Revolutionary Armed Colombian Forces (known by its Spanish acronym, FARC EP) and the national government. This body of work explores a wide range of social and economic outcomes, including potential impacts on ecotourism (Maldonado et al. 2018), deforestation (Prem, Saavedra, and Vargas 2020), violent crime (Peña and Dorussen 2021), health (Mora-García et al. 2025), firm creation (Bernal et al. 2024) and human capital accumulation for primary and secondary school (Prem, Vargas, and Namen 2023 and Montaña-Bardales and Palacios 2025).

Prem, Vargas, and Namen 2023 showed that the permanent ceasefire declared by FARC during peace negotiations with the Colombian government reduced school dropout and grade failure rates, increased high school graduation and test scores in areas most affected by FARC's presence. In addition, Montaña-Bardales and Palacios 2025 found that the peace agreement had a positive effect on gross enrollment, dropout rates, and academic performance, but gaps between the most affected municipalities and the least affected ones have widened.

However, to the best of my knowledge, there is no evidence of the effect at the higher education level. According to Ishiyama and Breuning 2012 and Milton and Barakat 2018, investment in education is essential to reduce the risk of conflict recurrence, particularly through the expansion of educational opportunities during the first five years after the end of conflict (Ishiyama and Breuning 2012).

Building on the evidence discussed above, it is plausible that access to higher education may also be influenced by violence reduction for three main reasons. First, reductions in violence improve security conditions across territories (Peña and Dorussen 2021; Perilla et al. 2024), allowing young people to complete secondary education (Montaña-Bardales and Palacios 2025) and continue their transition into higher education. Second, the previously documented improvements in academic performance strengthen the conditions for accessing higher education, as results from standardized national exams are a key requirement in

university admission processes. Third, a decline in violence may lead to a reallocation of public resources (Chen, Loayza, and Reynal-Querol 2008), reducing military spending and increasing social investment, particularly in education (Pais 2014). Such a shift can enhance access not only to primary and secondary schooling but also to higher education by expanding the availability of public resources and opportunities.

This study aims to fill these gaps by examining a central question: How does the reduction of violence following the Colombian peace agreement affect access to higher education? Understanding this causal relationship is crucial, as I discussed before, it shapes long-term development, social mobility, and the sustainability of peace.

2 Context

2.1 Armed conflict and peace agreement

The Colombian armed conflict is well-known for being the largest armed conflict in the Western Hemisphere (Justice for Colombia 2020). It formally started in 1964 with the creation of two guerrilla groups, FARC and National Liberation Army (known by its Spanish acronym, ELN), initially as an ideological conflict inspired by communism and the Cuban revolution. Over time, however, the rise of drug trafficking transformed the conflict, shifting its ideological motivation to economic interests surrounding drug trafficking and its profits (Justice for Colombia 2020; Richani 1997).

By the 1980's, a new actor entered the armed conflict: the United Self-Defense Forces of Colombia (AUC), commonly known as paramilitaries. This group was created to counter the guerrillas, promoting the economic interests of large landowners, businessmen, drug cartels, and the Colombian Army (Justice for Colombia 2020). Between 2002 and 2006, the AUC underwent a demobilization process, but internal divisions and uneven disarmament limited its success. Although the AUC formally ceased to exist, new criminal organizations emerged due to the rearmament of some demobilized members.

In 2012, the Colombian government started peace negotiations with the FARC group in Havana, Cuba. These talks led to a bilateral ceasefire in December 2014 and ended with the signing of the Final Peace Agreement in November 2016 (Prem, Vargas, and Namen 2023; Perilla et al. 2024).

According to the Commission for the Clarification of Truth, Coexistence, and Non-Repetition¹, established after the signing of the peace agreement, be-

¹Through Legislative Act 01 of 2017 and Decree 588 of 2017, the Commission for the Clarification of Truth, Coexistence and Non-Repetition was created, as a temporary and extrajudicial mechanism of the Comprehensive System of Truth, Justice, Reparation and

tween 1995 and 2018, 450,664 people lost their lives, 121,768 were forcibly disappeared, 50,770 were kidnapped, 16,238 children were recruited, and 752,964² suffered forced displacement. The FARC played a significant role in perpetrating these violent acts, which is why the peace agreement reached represented an important milestone in improving security conditions in the national territory (Prem, Vargas, and Namen 2023).

2.2 Higher education

The Colombian education system is composed of early education, preschool education, basic education (primary – 5 years and lower secondary – 4 years), upper secondary (2 years), and higher education. At the higher education level, students may pursue technical, technological, or university academic programmes at the undergraduate level, as well as postgraduate studies, including specializations, master’s degrees, and doctorates (MEN 2018b).

With the signing of the peace agreement, the Colombian State established a policy roadmap aimed at achieving peace and preventing the recurrence or escalation of armed conflict in the country. This roadmap includes the design and implementation of economic and social policies to guarantee citizens’ rights, with particular attention to the most vulnerable populations. Among the areas recognized as necessary to achieve this objective is education to promote social and rural development (Government and FARC-EP 2016).

Over the past two decades, the country has made significant efforts to expand coverage at this level of education. These efforts include policies to finance higher education demand through student-loan schemes, large-scale programs to support high-achieving students from low-income backgrounds, initiatives to strengthen the transition between secondary and higher education, and expanding the educational supply in rural areas (DNP 2014, DNP 2018, MEN 2018a, MEN 2017, DNP 2014).

Between 2014 and 2024, gross enrollment in higher education at the national level increased by 15.26%, increasing from 49.8% to 57.4%. This growth may be partly explained by improved security conditions in the country, which have facilitated the implementation of the programmes and policies described above.

Non-Repetition - SIVJRNR, to learn the truth of what happened in the framework of the armed conflict and contribute to the clarification of the violations and infractions committed during it, and offer a broad explanation of its complexity to the whole of society.

²These statistics are drawn from the analysis of 112 databases containing more than 8,775,884 records. According to the Commission for the Clarification of Truth, Coexistence, and Non-Repetition, the true figures could be as much as twice as high when sub-records are considered.

3 Research design

This study will follow a deductive approach. Based on human capital theory and on how it is affected by armed conflict and local security conditions, it aims to evaluate the effect of violence reduction, within the Colombian armed conflict context, on access to higher education.

The Colombian case is of particular interest given the long duration of the armed conflict, which has been widely studied for its negative effects on multiple social dimensions. However, the peace agreement and the subsequent decline in violence offer a unique opportunity to examine the recovery process of a society emerging from decades of conflict. This is relevant not only for supporting domestic public policies aimed at strengthening the factors that improve social conditions, but also for understanding the mechanisms that contribute to social reconstruction.

Furthermore, Colombia has public institutions that have carried out rigorous work in building databases that document the armed conflict and its territorial effects, as well as high-quality administrative and educational records. This makes it possible to run quantitative analysis in this type of context.

Regarding ethical considerations, this study relies on publicly available statistical information at the municipal level. The use of these data for research purposes is allowed by the sources and does not require specific permissions. Moreover, because the information is aggregated, it remains in line with good practices for handling sensitive data, and there is no risk of identifying individual information in databases.

To answer the research question, this study will adopt a causal inference approach, applying a difference-in-differences method. This method makes it possible to identify changes in access to higher education that can be attributed to violence reduction rather than to broader national trends, conditional on the validity of the method's underlying assumptions.

The following subsections describe in detail the data collection strategy and the analytical approach proposed to address the research question.

3.1 Data collection strategy

To evaluate the effect of violence reduction on access to higher education, I am going to use information at the municipal level. The dependent variable (outcome of interest) will be measured by the immediate transit rate to higher education for each municipality, available annually from 2013 to 2024. This indicator reflects access to and efficiency within the education system, as it measures the proportion of high school graduates who enter higher education programs (technical, technological, or professional) during the year immediately

following the completion of secondary education (MEN 2018b).

This information is publicly available through the Colombian Minister National Education - MEN. They are published online for each of the country's 32 departments, with separate files containing the information for all municipalities within each department. To construct a unified dataset, it is necessary to develop code that automates the download of all files, cleans and unifies tabular information, and merges data across years and municipalities.

Regarding violence, the Historical Memory Centre in Colombia is responsible for recovering, compiling, and analysing all data related to human rights violations committed within the armed conflict context. The Centre provides municipal-level information from 1996 to 2024 on the number of events and victims associated with targeted killings, terrorist attacks, kidnappings, sexual violence, massacres, enforced disappearances, antipersonnel mines, recruitment of children and adolescents, and other forms of violence.

The databases contain information on the alleged perpetrator, the date and location of the event, the type of violent act, the number of people affected, and other relevant variables. Using this information, I will construct a measure of violence equal to 1 for municipalities that were highly exposed to FARC-related violence, based on the distribution of the total number of FARC violent events relative to the population, recorded before the peace agreement signing. This measure aligns with standard approaches used in causal analyses that examine the relationship between conflict and education (Montaño-Bardales and Palacios 2025, Prem, Vargas, and Namen 2023, Unfried and Kis-Katos 2023).

However, other authors have used alternative measures of violence, including intensity indicators that capture multiple dimensions of the armed conflict and its territorial incidence (Suárez-Mesa and Gómez 2025), while others rely on the number of fatalities or victims (Diwakar 2015). The latter measure is not reliable in this study because a person may be victimized in multiple events, and in statistical data, it is not possible to identify such cases, which could lead to an overestimation of violence. Although this study primarily relies on the first approach, it will also incorporate alternative indicators, subject to data availability, to assess the robustness and sensitivity of the results.

Finally, it is important to consider municipal-level characteristics that reduce bias and control for structural differences across territories. I am going to complete the previous information with economic variables such as tax income, expenditure, transfers from the national government, and hectares cultivated with coca. These variables can affect the outcome, as municipalities with stronger economic conditions are generally better positioned to provide educational opportunities to their citizens. Moreover, there is evidence that larger areas planted with coca are associated with a greater school dropout rate in secondary school (Angulo 2024), which would imply an indirect effect on the

outcome of interest. This information is available at the municipal level through Colombian public entities, such as the National Planning Office and the Unit for the Comprehensive Attention and Reparation of Victims.

3.2 Data analysis strategy

Given the causal inference approach and the availability of longitudinal data, this study employs a quasi-experimental difference-in-differences method. This approach leverages the external variation generated by the signing of the peace agreement to compare changes in access to higher education between municipalities most affected by FARC presence (treated group) and those with limited or no FARC presence (control group), before and after the agreement.

The internal validity of this method relies on the assumption that, in the absence of the peace agreement, trends in access to higher education in treated municipalities would have evolved similarly to those in non-treated municipalities (parallel trends assumption). Besides, the method allows for controlling time-invariant unobservable factors that may influence the outcome.

To conduct the analysis and assess the key identification assumption, I will construct a panel dataset of 1,102 Colombian municipalities covering the period from 2013 to 2024 with the variables described in the data collection section.

Beyond the main results, I will also evaluate the sensitivity of the estimator to alternative measures of violence as a robustness check. This includes applying different cutoffs of the violence distribution to classify municipalities as highly affected by FARC presence—for example, using the median instead of the mean, or defining thresholds based on percentiles.

4 Contribution

A small body of work investigates the effects of post-conflict recovery on higher education. Most of the analyses consider only primary and secondary education. This study addresses this gap by evaluating the impact of violence reduction, followed by the signing of a peace agreement, on transit to higher education of youth from the territories more affected by the presence of the FARC guerrillas.

From a public policy perspective, the implementation of the agreements led the National Government to establish public policy strategies focused on the regions, promoting social inclusion and the enjoyment of rights through the provision of public goods and services, including education. This study contributes to providing empirical evidence on how peace-building processes can influence educational outcomes and whether these efforts translate into tangible results. It will offer valuable insights for policymakers seeking to design policies

that strengthen human capital formation in higher education level and promote inclusive development in post conflict societies.

5 Limitations

Due to the characteristics of the available information, the immediate transition indicator only includes young people who enter higher education in the year following the completion of secondary school. Therefore, it does not capture those who access higher education in later years. For this reason, the analysis reflects only part of the potential effect. Nevertheless, it can be taken as a lower-bound estimate of the effect, since any additional increase in access, unrecorded by the indicator, would imply a larger effect than the one estimated.

Furthermore, the peace agreement was signed only between the Colombian government and the FARC, one of the two guerrilla groups historically involved in the armed conflict. Although this agreement led to a substantial reduction in violence associated with that actor, the country continues to face violent dynamics driven by other armed groups, such as the ELN, as well as by criminal groups operating outside the armed conflict.

Hence, the analysis focuses exclusively on the reduction of violence in municipalities that were most affected by the presence and actions of the FARC. This means that the measure of violence reduction proposed in the analysis does not capture the full range of violent dynamics that persist across the territory. This may introduce potential bias in the estimated effect, as the observed changes in access to higher education could be influenced not only by the decline in FARC-related violence but also by the continuation or intensification of violence perpetrated by other armed groups.

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